TUTOR GUIDE • LEVEL 1 • TERM 1 • LESSON 9 OF 10

Draw and Measure



Big idea: Be exact to the nearest centimetre. A ruler counts centimetres. Start at the 0 mark, read the mark at the other end, and draw or measure a line exactly.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

1. At a glance

Strand	Length (Length in centimetres (cm); Measuring a line to the nearest cm; Drawing a line of a given length; Comparing lengths: how much longer) Strand label for the tutor only (MOE Measurement, length in cm, 1.1 to 1.4). It is not a claim that the program covers the school syllabus, and it must not appear on child or parent pages as 'MOE-aligned'. Deeper than the holiday cubes-and-ruler lesson, which is warm-up only. Term 1 goes beyond the school Primary 1 syllabus on purpose (owner decision, 22 Oct 2026): reading to the nearest cm, a line that starts at another mark, bent paths and the snail are enrichment, taught in class with a real ruler and pictures. Do not describe the program as MOE-aligned to families.
Objectives	• Use a ruler correctly: put the 0 mark at the start, keep it straight, read the mark at the other end. • Draw a straight line of a given whole number of centimetres, up to 14 cm. • Measure to the nearest centimetre: when the end is between two marks, choose the closer mark. • Compare two lengths and say how much longer one is. • Find the length of a line that starts at a mark other than 0, and of a bent path made of straight pieces.
Joining cold	**Remember warm-up covers:** Measure with same-size things laid end to end, no gaps and no overlaps, then count them; Find 0 on a ruler and count along the marks to 12; A centimetre (cm) is the gap from one mark to the next. Only counting to 20. A child who has never used a ruler is shown 0, the marks and the gaps with their own ruler in the warm-up. Nothing from earlier Term 1 lessons is needed. This lesson stands alone: it needs no earlier lesson except through the warm-up. It is one of the safer Term 1 lessons to miss (plan: the safest is Lesson 4). A child who misses it watches the recording and does workbook pages 1 and 2 with a ruler before Lesson 10.
Child materials	• A ruler with cm marks, starting at 0 (15 or 30 cm). No ruler? Tell your tutor before the lesson. • 3 or 4 things to measure; 10 paper clips (or small same-size things) • Workbook pages 1 to 5 (A4, printed at 100%), a pencil, a rubber, spare paper
Tutor materials	• A real ruler, a pencil and a strip of paper for your own use • A visible clock or a timer on your own device (the deck has no timer slide) Families keep a ruler at home, so no ruler is printed in the workbook. If a child has none, a grown-up can look for any straight-edged ruler or strip marked in centimetres, or tell the tutor before the lesson. Plastic or wooden rulers only, no metal edges. Nothing needs to be bought. No matches, toothpicks or anything sharp. Small things never go near the mouth or ears. Print the workbook at 100% (actual size, not 'fit to page'): the ruler pictures are real size, so a child can lay their own ruler beside them.
Tiers	Start here (6 questions, pages 1 and 2), Think harder (3, page 3), Go further (3, page 4), Investigation and look-back (page 5). Labels are neutral; nobody is told which page they are on.
Contest-style move	Used at Think harder only: a bent path: add the straight pieces; a line that starts at a mark other than 0 (end mark take away start mark); cutting a ribbon into pieces (the total stays the same). Never advertised to children or parents.

Before the lesson

- Solve every workbook question yourself with a real ruler first. Use something short, such as a crayon, for question 3.
- Tell parents a week before: a ruler marked in centimetres starting at 0, workbook pages 1 to 5 printed on A4 (there is no cut-out page this lesson), about 4 things to measure, 10 paper clips or small same-size things, a pencil and spare paper. Ask them to print the workbook at 100% (actual size), because the ruler pictures are real size.
- Have your own ruler, a pencil and a strip of paper ready. Draw on a shared whiteboard or a blank slide (not on paper under the webcam, which the recording cannot show). Have the slides open.
- Test the recording once with several tiles before the first live class: check that it captures the tutor, the shared screen and any unmuted voice, and not the children's faces. Write the setting you used in your tutor notes. If a family has no ruler, ask them before the lesson so a grown-up can find one; do not ask the child in front of the group.
- Say plainly if a parent asks: Term 1 goes beyond the school Primary 1 syllabus on purpose. Every Start here item is built to be reached with bars, bundles or a number line and a worked step, and the program does not claim to match any school syllabus. Items that need more than that sit at Think harder.

Safeguarding and recording (this lesson)

- Rulers stay flat on the table when not in use: no swinging, no flicking, no pointing. Plastic or wooden rulers only, no metal edges, no compasses. Small things stay on the table: nothing goes in the mouth or ears. No matches, toothpicks or anything sharp.
- This lesson uses the main room only: no breakout rooms. A parent sits nearby but does not give answers.
- Nobody holds paper or work up to the camera: a gallery view shows everyone. Children tick their own work against the answer slide, answer together on three, or show fingers or a thumb for a choice. There is no private chat and no board held up in front of the group. Rulers and things to measure stay on the child's own table.
- Never read out a child's name with a right or wrong answer. The recording captures the tutor, the shared screen and any voice that is not muted. Children may mute and keep their cameras off. Never show a child's page on the recording.
- A child who needs help ticks I am still stuck on their own look-back page. The grown-up may photograph that page and send it to the program's business WhatsApp after class (never to a personal phone); the tutor replies to the grown-up only. The next lesson starts with one short catch-up question for the whole group. No private messages to children and no private room.
- Movement break: children need a little clear space, soft shoes or socks on a safe floor, no stairs and no running. A child may stay seated and move only their arms.
- The look-back is never collected in front of the group or read out. A pass is always allowed.

2. Timing plan (90 minutes exactly)

Minutes	Part	Min	Slides	Script fills
0–4	**Opening**	4	1: Draw and Measure	3:03
4–10	**Remember warm-up**	6	3: Warm-up: measure with clips, Find 0 on your ruler, One centimetre	5:06
10–14	**Sneak-peek puzzles**	4	1: Sneak-peek puzzles	3:26
14–28	**Explore**	14	3: Explore: guess, then measure, Draw it!, What did we notice?	11:42
28–38	**Mini-lesson**	10	7: The parts of a ruler, Measure: three steps, Draw a line 7 cm long, Between two marks, A line that starts at 3, Your turn, Check: your turn	8:24
38–42	**Brain break**	4	1: Jump the tens, hop the ones	3:20

42–70	**Workbook**	28	5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page	26:39
70–76	**Talk time**	6	2: Talk and explain, Back to the sneak-peek	5:27
76–82	**Exit ticket**	6	2: Exit ticket, Check the exit ticket	5:08
82–85	**Look-back**	3	1: Look-back	2:34
85–90	**Flex time and goodbye**	5	1: Well done, inquirers!	1:11
	Total	**90**		**76:01**

The script needs 76:01 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 89:52. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 14 (end of sneak-peek), minute 42 (start of workbook), minute 70 (talk time). If more than 2 minutes behind at minute 42, skip in this order: In a slow class the workbook runs about 3 minutes over: take it from the 5-minute flex buffer and shorten choice time, not the exit ticket; the 12 cm line in Explore round 2; optional items and extra quick questions; the mini-lesson preview of the closer mark; the optional slide. Never skip the look-back or the camera and privacy moments.

3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

0:00 – 4:00 Opening

script 3:03 of 4:00

0:00	SLIDE	Show slide **Draw and Measure** (s-title)
0:25	REC	*If you are watching the recording: hello! Get a pencil, paper, your workbook and your ruler. Pause or rewind any time.*
0:36	SAY	Hello, inquirers! Pencil ready, and your ruler on the table. Cameras can be on or off, as you like.
0:45	SAY	This lesson is recorded so a friend who is away can catch up. The recording has me, the shared screen and any voice that is not muted. You may keep your microphone and camera off.
1:02	SAY	Microphone off until I ask you to speak. To answer, say it with me on three, show fingers, or give a thumbs up. Nobody holds paper up to the camera.
1:18	DO	Greet the whole group warmly. Do not use children's first names on the recording. Ask everyone to put a finger on the 0 mark of their ruler and say zero quietly, so you can hear it. Remind them: a ruler stays flat on the table, and it is only for measuring. [0:40]
1:58	SAY	Here is a question to start. How long is a pencil? Show me with your two hands, as far apart as you think. Hold still.
2:10	CHILDREN	Children show a guess with their hands. Say what you see in general (some wide, some narrow). Nobody is right or wrong yet. [0:35]
2:45	SAY	Hands are not very exact. My hands and your hands are not the same size. Today we learn to measure and to draw lines exactly, so that everybody gets the same answer.
3:01	SAY	Let us warm up.
	NOTE	Ruler check: if a family has no ruler, a grown-up can look for any straight-edged ruler or strip marked in centimetres, or tell the tutor before the lesson; do not ask a child in front of the group. Keep the greeting short and warm. Say the recording statement and the microphone routine in the same words every lesson.

4:00 – 10:00 Remember warm-up

script 5:06 of 6:00

4:00	SLIDE	Show slide **Warm-up: measure with clips** (s-rem1)
4:25	REC	*Recording: join in! Lay clips along a pencil, find 0 on your ruler, and count the marks out loud. Pause if you need longer.*
4:37	SAY	Get your pencil and your paper clips, or small things that are all the same size. Lay them end to end along your pencil. No gaps. No overlaps. Then count them.
4:52	CHILDREN	Children lay clips (or pasta, short straws) along a pencil and count them. [0:50]
5:42	ASK	How many clips long is your pencil? Say your number or show it on your fingers. [wait about 20 s for answers]
6:10	SAY	My pencil is 6 big clips long. Now I use small clips: 9. Same pencil, different numbers! The clips are not the same size. So people agree to use one unit that is the same for everybody: the centimetre.
6:30	SLIDE	Show slide **Find 0 on your ruler** (s-rem2)
6:55	SAY	Now your ruler. Find the 0 mark. Put your finger on it. Count along the marks with me: 0, 1, 2, 3, all the way to 10.
7:08	CHILDREN	Count the marks together, pointing with a finger, from 0 to 10, then on to 12. [0:30]
7:38	ASK	Which mark comes just after 4? [wait about 10 s for answers]

7:52	**ASK**	Which mark comes just before 10? [wait about 10 s for answers]
8:04	**SLIDE**	Show slide **One centimetre** (s-rem3)
8:30	**SAY**	The space between two marks is one centimetre. We write it cm. From 0 to 1 is 1 cm. Slide your finger from 0 to 5. Five gaps: that is 5 cm.
8:46	**ASK**	Slide your finger from 0 to 8. How many centimetres is that? [wait about 15 s for answers]
	IF STUCK	A child who counts the marks instead of the gaps (says 6 for 0 to 5): put a finger on the 0 and say zero, then count each jump aloud: one, two, three, four, five. The 0 is where we start, it is not a centimetre.
	NOTE	Answers: any clip count is fine (the number depends on the clips). Then 5, 9, 8. Children answer with fingers, thumbs or a whispered number. Nobody writes. A child who has never used a ruler sees the 0 mark and the gaps on their own ruler first.

10:00 – 14:00 Sneak-peek puzzles

script 3:26 of 4:00

10:00	**SLIDE**	Show slide **Sneak-peek puzzles** (s-sneak)
10:25	**REC**	*Recording: pause now and try both puzzles on your own paper. Guessing is fine. Press play for the next part.*
10:35	**SAY**	Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked.
10:48	**READ**	Puzzle one. Draw a dot. Use your ruler to draw a line from the dot, exactly 3 cm long.
10:58	**CHILDREN**	Children draw the dot and the 3 cm line on their own paper. [0:55]
11:53	**READ**	Puzzle two. Ali puts one end of a pencil at the 1 mark of his ruler. The other end is at the 9 mark. Ali says it is 9 cm long. Is he right? Write yes or no. Then write why.
12:15	**CHILDREN**	Children write or draw their answer. [0:55]
13:10	**SAY**	Thank you. Keep your paper and your own answer: nobody shows it. We will come back to puzzle two in a few minutes, and you can check your own thinking then.
	NOTE	Puzzle 1 answer: a line 3 cm long (the 0 mark at the dot, the 3 mark at the other end). Puzzle 2 answer: no, 8 cm, because the pencil starts at the 1 mark. Look for good ideas to praise later (a child who started at 0, or who noticed the 1). Do not write scores or keep a list.

14:00 – 28:00 Explore

script 11:42 of 14:00

14:00	**SLIDE**	Show slide **Explore: guess, then measure** (s-explore1)
14:25	**REC**	*Recording: pause now. Guess, then measure three things. Then draw lines of 3, 8 and 12 centimetres. Press play when you have tried.*
14:36	**SAY**	Today your ruler is your teacher. Find three things: a pencil, a spoon, a rubber or a toy. First guess how many centimetres long each one is. Then measure.
14:51	**SAY**	Put one end at the 0 mark, not at the edge of the ruler. Read the mark at the other end. If the end is between two marks, either mark next to it is fine today. Choosing the closer mark is a Think harder idea.
15:14	**CHILDREN**	Round 1: each child guesses, then measures three things and shows a thumbs up when done. Tutor listens for the 0 mark in the answers: start at 0. [3:50]
19:04	**ASK**	Tell me about one of your things. How long was it? How close was your guess? [wait about 50 s for answers]
20:02	**SLIDE**	Show slide **Draw it!** (s-explore2)
20:26	**SAY**	Round two: draw it! Put a dot. Put the 0 mark on the dot. Make a small mark at the end. Join the dot and the mark with a straight line. First, draw a line 3 cm long.
20:46	**CHILDREN**	Children draw 3 cm. [0:35]
21:20	**SAY**	Now draw a line 8 cm long.

21:24	**CHILDREN**	Children draw 8 cm. [0:40]
22:04	**SAY**	Now a line 12 cm long.
22:07	**CHILDREN**	Children draw 12 cm. [0:45]
22:52	**CHILDREN**	Children look at their own three lines: does each one start at the 0 mark? Thumbs up when checked. No paper is shown to the camera. [0:20]
23:12	**ASK**	Ravi started his line at the very edge of his ruler, not at the 0 mark, and stopped at the 8 mark. Is his line exactly 8 cm? How do you know? [wait about 40 s for answers]
24:08	**SLIDE**	Show slide **What did we notice?** (s-explore3)
24:33	**ASK**	What did you notice? Where do we start measuring? Why do we all use centimetres? [wait about 45 s for answers]
25:26	**SAY**	We start at the 0 mark. We read the mark at the other end. If the end is between two marks, we say about, and later we learn to choose the closer one.
	IF STUCK	A child who measures from the edge of the ruler: ask them to find the 0 mark and slide the thing along until its end is on the 0. A child with a 15 cm ruler measuring something longer: ask them to measure a shorter thing, or mark the end of the ruler with a finger and slide on.
	IF STUCK	A child with nothing to measure: any straight line works, such as the edge of a book or a line they draw on paper.
	NOTE	Main room only: no breakout rooms in this lesson. Pairs are optional: a grown-up may check the 0 end. Ravi answer: no. He began a little before the 0 mark, so his line is a little longer than 8 cm. Praise the reason, not the speed.

28:00 – 38:00 Mini-lesson

script 8:24 of 10:00

28:00	**SLIDE**	Show slide **The parts of a ruler** (s-m1)
28:25	**REC**	*Recording: keep watching and copy each example with your own ruler. Rewind if you miss a step.*
28:34	**SAY**	Look at your ruler. The numbers sit on the marks. 0 is where we start. The gap from one mark to the next is one centimetre.
28:46	**ASK**	How many centimetres from the 0 mark to the 6 mark? Slide your finger and say it. [wait about 12 s for answers]
29:07	**SLIDE**	Show slide **Measure: three steps** (s-m2)
29:32	**SAY**	To measure: one end at 0, the ruler straight, and read the mark at the other end. Watch my pencil.
29:42	**DO**	Point to the pencil on the slide and say the three steps. Say each step as you show it. Read the mark at the tip aloud with the unit. [0:30]
30:12	**SLIDE**	Show slide **Draw a line 7 cm long** (s-m3)
30:37	**SAY**	To draw a line 7 cm long: a dot for the start. The 0 mark on the dot. A small mark at the 7. Then join them with a straight line. I draw while you watch.
30:55	**DO**	Draw a 7 cm line on a shared whiteboard or a blank slide, saying each step (the slide shows the finished line). Check at the end: 0 at the start, 7 at the end. [0:40]
31:35	**SLIDE**	Show slide **Between two marks** (s-m4)
32:00	**SAY**	A little extra for Think harder. Sometimes the end is not on a mark. Here it is between 8 and 9, and closer to 9. We say about 9 centimetres: the nearest centimetre.
32:16	**ASK**	Another one: between 12 and 13, and closer to 12. How long, to the nearest centimetre? [wait about 15 s for answers]
32:40	**SLIDE**	Show slide **A line that starts at 3** (s-m5)
33:04	**SAY**	A line that does not start at 0. It starts at the 3 mark and ends at the 8 mark. Count the jumps: 1, 2, 3, 4, 5. It is 5 cm long, not 8!

33:22	**ASK**	Why is it 5 cm and not 8 cm? [wait about 20 s for answers]
33:46	**SLIDE**	Show slide **Your turn** (s-m6)
34:12	**READ**	Your turn. Draw a line 5 cm long. Then: a line goes from the 2 mark to the 9 mark. How long is it? Write both answers on your paper.
34:28	**CHILDREN**	Children draw 5 cm and work out the second line. Wait until most have stopped writing. [1:15]
35:43	**SLIDE**	Show slide **Check: your turn** (s-m6a)
36:08	**READ**	The 5 cm line starts at 0 and ends at 5. The other line is 7 cm: count 7 jumps from 2 to 9, or take 2 from 9.
	NOTE	Key words every time: ruler, mark, centimetre (cm), start at 0. Nearest is a Think harder word. Do not teach millimetres or metres. Do not call the 0 a centimetre. The jump idea (count the gaps) is the same as hopping along a number line.
	IF STUCK	If a child draws from the end of the ruler: ask them to put a finger on the 0 mark and slide the ruler until the 0 is on the dot. If a child says the 2-to-9 line is 9 cm: ask them to count the jumps from the 2.

38:00 – 42:00 Brain break

script 3:20 of 4:00

38:00	**SLIDE**	Show slide **Jump the tens, hop the ones** (s-break)
38:25	**REC**	*Recording: stand up and join in! Big jump for ten centimetres, small hop for one.*
38:32	**SAY**	Stand up and get a sip of water. We move our bodies today. Pretend a big jump is ten and a small hop is one. So 14 is one big jump and four small hops.
38:50	**CHILDREN**	Call 14, 23, 30, 8. Children jump the tens and hop the ones. Say each count clearly; children count quietly along. [2:30]

42:00 – 70:00 Workbook

script 26:39 of 28:00

42:00	**SLIDE**	Show slide **Workbook page 1: Start here** (s-ws1)
42:25	**REC**	*Recording: pause now and do workbook page 1 with your ruler. When you finish a page, press play to check it.*
42:36	**READ**	Page 1, Start here. Question 1: in the box, draw a line 7 centimetres long. Start it at the 0 mark. Question 2: draw a line 9 cm long. Ali's line is 4 cm long. How much longer is your line? Question 3: measure something short from your table, such as a crayon or a rubber. Write your guess first. Then measure it in whole cm.
43:12	**CHILDREN**	Page 1, questions 1 to 3. Tutor looks around the gallery and asks how do you know. Wait 10 seconds before helping. Move to the answer slide only when most children have stopped writing; anyone not finished keeps going and checks later. [9:00]
52:12	**SLIDE**	Show slide **Check page 1** (s-ws1ans)
52:36	**READ**	Answers for page 1. If you are still working, carry on and check later. Question 1: your line ends at the 7 mark. Question 2: your line is 5 cm longer than Ali's. Question 3: any answer close to your guess is fine. Check your own page in another colour.
53:04	**SAY**	After question 3 you may choose: carry on, or try Think harder. Everyone may choose, at any time.
53:13	**SLIDE**	Show slide **Workbook page 2: Start here** (s-ws2)
53:38	**READ**	Page 2. Question 4: look at the ruler. How long is each pencil, and how much longer is Ali's? Question 5: Mei's line is 5 cm long. Draw a line 3 cm longer. Question 6: look at Mei's line on the ruler. Mei says it is 8 cm long. Is she correct? Circle Yes or No. Then write how long her line is.
54:12	**CHILDREN**	Page 2, questions 4 to 6. Move to the answer slide only when most children have stopped writing. [9:00]
63:12	**SLIDE**	Show slide **Check page 2** (s-ws2ans)
63:37	**READ**	Answers for page 2. If you are still working, carry on and check later. Question 4: 13 cm, 9 cm, and 4 cm longer. Question 5: 8 cm. Question 6: no, her line is 7 cm.

63:57	SLIDE	Show slide **Choose your next page** (s-choice)
64:22	SAY	When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time.
64:34	CHILDREN	Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. [4:05]
	IF STUCK	Q1, Q2 and Q5 (drawing): put a finger on the 0 mark, slide the ruler to the dot. Q3: guess first, then measure; a guess that is far off is fine. Q4 and Q6: the ruler picture is printed at real size, so children may lay their own ruler beside it. Q6: ask where her line ends on the ruler.
	IF STUCK	A child who is still stuck ticks I am still stuck on their own look-back page. The grown-up may photograph that page and send it to the business WhatsApp after class. Open the next lesson with one short catch-up question for the whole group. Never a private message to a child, never a private room.
	NOTE	Think harder is for ideas taught today in a new setting. Go further is for mastery and extension. Do not announce who is on which page. Pairs are never split unless both children agree.

70:00 – 76:00 Talk time

script 5:27 of 6:00

70:00	SLIDE	Show slide **Talk and explain** (s-talk)
70:25	REC	*Recording: pause. Tell a grown-up your answer to the talk question, and say why.*
70:32	READ	Mei measures a pencil and says it is 15 cm. Ben measures the same pencil and says it is 16 cm. Why might their answers be different? How can we check?
70:49	CHILDREN	Think for a moment. Put a pencil next to your ruler and show your grown-up what you would do. [0:50]
71:39	CHILDREN	Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [1:00]
72:39	ASK	Who would like to share? Volunteers first. A pass is always fine. [wait about 90 s for answers]
74:15	SLIDE	Show slide **Back to the sneak-peek** (s-sneakans)
74:40	SAY	Remember puzzle two? Ali put his pencil at the 1 mark and the other end was at the 9 mark. He said 9 cm. The pencil is 8 cm long, because it starts at the 1, not at the 0.
75:00	ASK	Did anyone change their mind about puzzle two while we worked? You can just nod. [wait about 20 s for answers]
	NOTE	Model answer: the tip may be between the 15 and the 16 mark and the two children chose different marks, or one child did not start at the 0 mark. Check by measuring again with the 0 at the start. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed.

76:00 – 82:00 Exit ticket

script 5:08 of 6:00

76:00	SLIDE	Show slide **Exit ticket** (s-exit)
76:25	REC	*Recording: pause and do the exit ticket on your own, with your ruler. Then press play to check it.*
76:34	SAY	Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5. You may use your ruler. I will read them to you.
76:48	READ	Question 1: in the box, draw a line 6 centimetres long. Start at the 0 mark. Question 2: Siti's rubber ends at the 6 mark and her pencil ends at the 15 mark. Both start at the 0 mark. How many cm longer is the pencil? Question 3: a line starts at the 3 mark and ends at the 10 mark. How long is the line?
77:24	CHILDREN	Children answer E1 to E3 on their own. [2:30]
79:54	SLIDE	Show slide **Check the exit ticket** (s-exitans)
80:19	READ	Question 1: your line starts at 0 and ends at 6. Question 2: 9 centimetres longer. Question 3: 7 centimetres, because 10 take away 3 is 7.
80:34	SAY	Tick the ones you got right, in a different colour. Well done. Show your ticket to a grown-up.

80:43	**ASK**	Show me with your fingers how many you got right: three, two, one, or none yet. A pass is fine. [wait about 15 s for answers]
	NOTE	Look at the room as a whole, not at any one tile. If many children show one or two fingers, spend two minutes at the start of Lesson 10 on starting at the 0 mark and on lines that start at another mark, in the open session, no names. Do not write a child's name next to an answer and do not read out anyone's result.

82:00 – 85:00 Look-back

script 2:34 of 3:00

82:00	**SLIDE**	Show slide **Look-back** (s-look)
82:25	**REC**	*Recording: pause and fill in your look-back box. You can pass.*
82:30	**SAY**	Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. If you still need help, tick I am still stuck and show your grown-up. You can pass. Nobody reads it out.
82:54	**CHILDREN**	Children fill in the look-back box on page 5. [1:40]

85:00 – 90:00 Flex time and goodbye

script 1:11 of 5:00

85:00	**SLIDE**	Show slide **Well done, inquirers!** (s-end)
85:25	**REC**	*Recording: well done for finishing! Show a grown-up one thing you found out.*
85:32	**SAY**	Well done, inquirers. Next lesson is Story Steps, the last lesson of the term, with Mi's Challenge Quest, a game with Mi. You need about 20 counters, such as buttons or coins, your ruler, a pencil and paper, and the Quest Log in its envelope, kept closed. You also need a quest partner: a grown-up, or a brother or sister. Your grown-up will get a note about the Quest Log to print.
	SLIDE	Show slide **Go further: how many cuts?** (s-opt) *(optional)*
	SAY	Optional, only if you are on time: a strip of paper is 10 cm long. You may fold it at the 1 mark, the 2 mark and so on, but not at the ends. How many places can you fold? And if the strip is 20 cm long?
	CHILDREN	Short think time, then take two answers. [1:30]
	SLIDE	Show slide **If we have time: pick one** (s-menu) *(optional)*
	SAY	Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each).
	NOTE	This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: 9 places for the 10 cm strip, 19 for the 20 cm strip (one fewer than the length, because we cannot fold at the two ends). If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on.
86:08	**SAY**	Goodbye, and thank you to the grown-ups.

3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content. If you are more than 3 minutes ahead at minute 42, use one menu activity then.
- A few children ahead: they may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask "How do you know?" and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide.

Activity	Minutes	You say	Recording cue	Answer

Is it always true?	4 (optional)	Mei says a line that ends at the 9 mark is always 9 cm long. Is that always true? Draw lines that end at the 9 mark and start at 0, at 2 and at 4. Measure each one.	*Recording: pause. Draw 3 lines that all end at the 9 mark: one starts at 0, one at 2, one at 4. How long is each?*	Not always. A line that starts at 0 and ends at 9 is 9 cm. Starting at 2 it is 7 cm, and starting at 4 it is 5 cm. It is only true when the line starts at the 0 mark.
Make a puzzle for your grown-up	4 (optional)	Draw a path with 3 straight pieces. Write the total length on the back, but not the pieces. Give it to your grown-up to measure each piece and add them. Do they get your total?	*Recording: pause. Draw a path with 3 straight pieces. Ask a grown-up to measure and add them. Does it match your total?*	No single answer. The measured pieces should add up to the total on the back, give or take 1 cm.
Teach it back	3 (optional)	Pretend your grown-up or a toy has never used a ruler. Teach them how to draw a line 9 cm long, step by step, and say why the 0 mark goes on the dot.	*Recording: pause. Teach a grown-up how to draw a line 9 cm long with a ruler, in steps.*	A dot for the start. The 0 mark on the dot. A small mark at the 9. Join them with a straight line. Check: 0 at the start, 9 at the end. We start at 0 because the centimetres are counted from the 0 mark.

4. Questions, answers, hints and common mistakes

Sneak-peek puzzles (minutes 10 to 14)

Puzzle 1 (access): Draw a dot. Use your ruler to draw a line from the dot, exactly 3 cm long.

> **Answer: A line from the dot that is 3 cm long: the 0 mark on the dot and the 3 mark at the other end.**

Puzzle 2 (think): Ali puts one end of a pencil at the 1 mark of his ruler. The other end is at the 9 mark. Ali says it is 9 cm long. Is he right? Write yes or no. Then write why.

> **Answer: No. It is 8 cm long. The pencil starts at the 1 mark, not the 0 mark, so the first centimetre of the ruler is not part of the pencil (9 take away 1 is 8).**

Unmarked. Never put answers on boards in front of the group and never read a name with an answer. Reveal the answers where your script says (usually Talk time).

Start here (workbook pages 1 and 2)

S1 **In the box, draw a line 7 cm long. Start at the 0 mark.**

Your line ends at the ☐ mark.

> **Answer: The line ends at the 7 mark (it starts at the 0 mark and is 7 cm long)**

Hint (say this first): Make a dot. Put the 0 mark on the dot. Make a small mark at the 7. Join them with a straight line.
Worked solution: Put a dot for the start. Line up the 0 mark with the dot. Make a small mark at the 7. Join the dot and the mark with a straight line. The line ends at the 7 mark, so it is 7 cm long.
Common mistakes: Starting at the edge of the ruler instead of the 0 mark (the line comes out a little too long).; Drawing to the 8 mark because the 0 was counted as one.
Time for a middle child: about 2.5 min.

S2 **In the box, draw a line 9 cm long. Ali's line is 4 cm long.**

Your line is ___ cm longer than Ali's line.

> **Answer: 5 cm longer (9 take away 4)**

Hint (say this first): Draw your line from the 0 mark to the 9 mark. To see how much longer, count the cm from 4 to 9.
Worked solution: Your line is 9 cm and Ali's line is 4 cm. Count on from 4 to 9: that is 5 jumps. So your line is 5 cm longer (9 take away 4 is 5).
Common mistakes: Writing 13 by adding the two lengths.; Starting the 9 cm line at a mark other than 0.; Writing 9 (the length of the line you drew).
Time for a middle child: about 3 min.

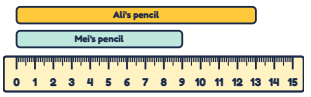
S3 **Measure something short from your table, such as a crayon or a rubber. Write your guess first. Then measure it in whole cm.**

(a) My guess: ___ cm
(b) I measured: ___ cm

> **Answer: Any answer that matches the thing you measured and is close to your guess. A grown-up checks.**

Hint (say this first): Put one end at the 0 mark. Keep the ruler straight. Read the mark at the other end. If the end is between two marks, either mark next to it is fine today.
Worked solution: A grown-up checks: put one end on the 0 mark and read the mark at the other end. A crayon is about 8 to 9 cm and a rubber about 5 to 6 cm. Accept any whole number within 1 cm of the tip.
Common mistakes: Starting at the edge of the ruler instead of the 0 mark.; Writing the guess in the measure space, or leaving off the unit.
Time for a middle child: about 3 min.

S4 **Look at the ruler. Ali's pencil and Mei's pencil both start at the 0 mark.**



(a) Ali's pencil is ___ cm long.
(b) Mei's pencil is ___ cm long.
(c) Ali's pencil is ___ cm longer than Mei's.

Answer: 13 cm; 9 cm; Ali's is 4 cm longer

Hint (say this first): Both pencils start at 0, so the mark at the tip is the length. To find how much longer, count on from 9 to 13.
Worked solution: Both start at the 0 mark, so each tip mark is the length: Ali 13 cm, Mei 9 cm. From 9 to 13 is 4 jumps (13 take away 9 is 4), so Ali's pencil is 4 cm longer.
Common mistakes: Adding 13 + 9 = 22.; Writing 9 as the difference (that is the length of Mei's pencil).
Time for a middle child: about 3 min.

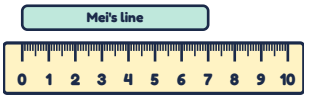
S5 **Mei's line is 5 cm long. In the box, draw a line 3 cm longer than Mei's line.**

Your line is ___ cm long.

Answer: The second line is 8 cm long (5 + 3)

Hint (say this first): Your line is Mei's line and 3 more. How long is that? Draw it with the 0 mark at the start.
Worked solution: Mei's line is 5 cm. 3 cm longer is 5 + 3 = 8 cm. Draw it from the 0 mark to the 8 mark.
Common mistakes: Drawing a line 3 cm long.; Starting the line at the edge of the ruler, so it is a little too long.
Time for a middle child: about 3 min.

S6 **Look at Mei's line on the ruler. Mei says it is 8 cm long.**



(a) Is Mei correct? Circle Yes or No.
(b) Her line is ___ cm long.

Answer: No. Her line is 7 cm. She should draw on to the 8 mark

Hint (say this first): Where does her line start and end on the ruler? Read the end mark.
Worked solution: Mei started at 0 and stopped at the 7 mark, so her line is 7 cm long. A line 8 cm long must end at the 8 mark. She should draw on to the 8 mark.
Common mistakes: Saying yes because Mei said 8.; Counting the 0 as one, so the 7 mark looked like 8.
Time for a middle child: about 2.5 min.

Think harder (page 3): the taught idea in a new setting

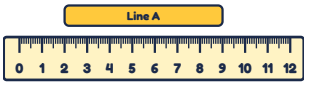
T1 **Use your ruler to draw a path with 3 straight pieces: 5 cm, 3 cm and 4 cm.**

The whole path is ___ cm long.

Answer: 12 cm (5 + 3 + 4)

Hint (say this first): Draw one piece at a time. Turn a corner and draw the next piece. Then add the three lengths.
Worked solution: Draw 5 cm, turn, draw 3 cm, turn, draw 4 cm. The path is as long as all three pieces together: 5 + 3 = 8, and 8 + 4 = 12 cm.
Common mistakes: Measuring in a straight line from the start to the end of the path.; Drawing one long line instead of turning at the corners.
Time for a middle child: about 5 min.

T2 **Look at line A on the ruler. Line B starts at the 3 mark and is 6 cm long.**

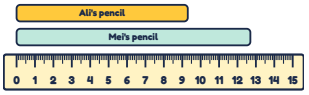


(a) Line A is ___ cm long.
(b) Line B ends at the ___ mark.

Answer: 7 cm; the 9 mark

Hint (say this first): Count the jumps from the start mark to the end mark. For line B, jump 6 along from the 3 mark.
Worked solution: (a) From the 2 mark to the 9 mark is 7 jumps: 9 take away 2 is 7 cm, not 9 cm. (b) Start at the 3 mark and jump on 6: 3 + 6 = 9. The line ends at the 9 mark.
Common mistakes: Writing 9 for line A because the end is at the 9 mark.; Taking 6 away from 3, or writing 6 for line B.
Time for a middle child: about 4 min.

T3 **Look at the ruler. Both pencils start at the 0 mark. Each tip is between two marks.**



(a) To the nearest cm, Ali's pencil is ___ cm long.
(b) To the nearest cm, Mei's pencil is ___ cm long.
(c) Mei's pencil is ___ cm longer than Ali's.

Answer: 9 cm; 13 cm; 4 cm longer

Hint (say this first): For each pencil, look at the two marks around the tip. Choose the mark the tip is closer to. Then count on from Ali's length to Mei's.
Worked solution: Ali's tip is just past the 9 mark, so it is closer to 9: 9 cm. Mei's tip is just before the 13 mark, so it is closer to 13: 13 cm. From 9 to 13 is 4 jumps, so Mei's pencil is 4 cm longer.
Common mistakes: Choosing the mark the tip has not reached (12 for Mei).; Writing 10 for Ali because the tip looks nearly there.; Writing 3 by using 12 and 9.
Time for a middle child: about 5 min.

Go further (page 4): mastery and extension

G1 **A ribbon is 30 cm long. Mei cuts all of it into 3 pieces: A, B and C. Piece A is 8 cm long. Piece B is 3 cm longer than piece A. How long is piece C?**

Piece C is ___ cm long.

Answer: Piece C is 11 cm (B is 11 cm, A and B together are 19 cm, and 30 take away 19 is 11)

Hint (say this first): Find piece B first. Then add A and B. How much of the 30 cm is left for C?
Worked solution: B is 8 + 3 = 11 cm. A and B together are 8 + 11 = 19 cm. The ribbon is 30 cm, so C is 30 take away 19, which is 11 cm. Check: 8 + 11 + 11 = 30.
Common mistakes: Taking 30 - 8 = 22 and stopping.; Writing 3 for piece B instead of 3 more than A.
Time for a middle child: about 5 min.

G2 **A snail crawls along a long ruler from the 4 mark to the 13 mark. Then it crawls back 5 cm.**



(a) The snail is now at the ___ mark.
(b) The snail has crawled ___ cm in all, counting the way back.

Answer: At the 8 mark; it crawled 14 cm in all (9 cm forward and 5 cm back)

Hint (say this first): Forward: from 4 to 13 is how many cm? Back: take 5 from the 13 mark. For the total, add the way forward and the way back.
Worked solution: Forward from the 4 mark to the 13 mark is 9 cm. Back 5 cm: 13 take away 5 puts the snail at the 8 mark. In all it crawled 9 + 5 = 14 cm, although it is only 4 cm from where it began.
Common mistakes: Writing 9 for the total and forgetting the way back.; Writing 4 as the mark (the distance from the start, not the mark).
Time for a middle child: about 6 min.

G3 **Draw a path with 3 straight pieces. The pieces are all different lengths, in whole cm. The whole path is exactly 20 cm long. Many answers are right.**

My 3 lengths: ___ cm, ___ cm and ___ cm.

Answer: Many answers are right: any 3 different whole numbers that add up to 20, for example 5, 7 and 8 (5 + 7 + 8 = 20).

Hint (say this first): Pick 3 different numbers and add them. Too small? Make one longer. Too big? Make one shorter. Then draw each piece.
Worked solution: One answer is 5, 7 and 8: 5 + 7 is 12 and 12 + 8 is 20. Other answers are 3, 8 and 9, or 4, 7 and 9. If the total is not 20, change one piece until it is.
Common mistakes: Two pieces the same length (for example 5, 5 and 10).; A total of 19 or 21 because a piece was drawn a little too long or too short.
Time for a middle child: about 7 min.

Investigation (page 5): Does the start mark matter?

Easy first step: Draw a line 6 cm long from the 2 mark. Where does it end?
Then: Try 3 more start marks, 6 cm each time. Lines 1 to 3: write the start and end marks. Line 4: I notice ...

> **Model answer: The length stays 6 cm every time. From the 2 mark the line ends at 8; from 3 it ends at 9; from 5 it ends at 11. The end mark is always the start mark plus 6, so the end mark take away the start mark is always 6. Reading only the end mark gives the length only when the line starts at 0.**

Look for: A child who tries more than one start mark, writes both marks, and says that the end mark take away the start mark (or the number of jumps) is the length. A 15 cm ruler allows start marks up to 9.
Common mistakes: Writing the end mark as the length (8 for a line from 2 to 8).; Trying only one start mark.

Talk and explain (minutes 70 to 76)

Mei measures a pencil and says it is 15 cm. Ben measures the same pencil and says it is 16 cm. Why might their answers be different? How can we check? Sentence starter: I think ... because ...

> **Model answer: The tip may be between the 15 mark and the 16 mark, so one child chose the other mark. Or one child did not start at the 0 mark. We can check by measuring again with the 0 mark at the start, and by choosing the closer mark.**

Look for: A reason about the start at 0 or about an end between two marks, and a way to check (measure again). Praise the check, not who is right. Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

Exit ticket (minutes 76 to 82): 3 questions

E1 **In the box, draw a line 6 cm long. Start at the 0 mark.**
Your line ends at the ☐ mark.

> **Answer: A line from the 0 mark to the 6 mark**

Worked solution: Put the 0 mark of the ruler on the dot and draw to the 6 mark. The line ends at the 6 mark.
Common mistakes: Starting at the edge of the ruler.; Ending at the 7 mark after counting the 0 as one.

E2 **Siti's rubber ends at the 6 mark and her pencil ends at the 15 mark. Both start at the 0 mark.**
The pencil is ___ cm longer than the rubber.

> **Answer: 9 cm longer**

Worked solution: The rubber is 6 cm and the pencil is 15 cm. 15 take away 6 is 9, so the pencil is 9 cm longer.
Common mistakes: Adding: 21.; Writing 15 (the pencil, not the difference).

E3 **A line starts at the 3 mark and ends at the 10 mark. How long is the line?**
The line is ___ cm long.

> **Answer: 7 cm. 10 take away 3 is 7 (7 jumps from 3 to 10)**

Worked solution: Count the jumps from 3 to 10: 7 jumps. Or take 10 take away 3, which is 7. The line is 7 cm long, not 10 cm.
Common mistakes: Writing 10 (the end mark).; Writing 13 by adding.

Look-back (minutes 82 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (starting at the 0 mark; drawing a line exactly; a line that starts at another mark; adding the pieces of a path; something else); *What helped me* (my ruler; drawing it; trying again; I asked for help; I was not stuck); *Next time I will try* (drawing it; using objects; checking my answer; asking a question); *I am still stuck (my grown-up will photograph this page for my tutor)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

5. Home practice and online practice

#	Tier	Question	Answer

H1	Start here	Draw a line 8 cm long on spare paper. Start it at the 0 mark of your ruler. Then check: which mark does it end at?	It starts at 0 and ends at 8
H2	Start here	A pencil starts at the 0 mark and its tip is at the 17 mark. A crayon is 11 cm long. How many cm longer is the pencil?	6 cm
H3	Start here	Measure a spoon, a toy and a crayon in whole cm. Write the 3 lengths in order, shortest first.	Many answers are right. Check one by measuring again.
H4	Think harder	A line starts at the 4 mark and ends at the 15 mark. How long is the line? Write: I know because …	11 cm (15 take away 4)
H5	Go further	Draw a path with 3 straight pieces. The pieces are all different lengths. The whole path is 25 cm long. Write the 3 lengths. A grown-up checks with a ruler.	Many answers are right: any 3 different whole numbers that add up to 25, for example 6 + 8 + 11 = 25.

Online practice (`out/l09/practice`, key `t1l09`): 13 items; 10 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

6. Parent line for the term report

Your child is learning to measure and draw lines in centimetres with a ruler and to say how they know. This week: starting at 0, reading the end mark, and comparing lengths.

Term line (plan, section 5): Your child is learning to see numbers as tens and ones, to notice patterns and to say why. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

7. Slide list

#	Time	Slide	On screen	Speaker note
1	0–4	**Draw and Measure** s-title	TODAY WE WILL: / Use a ruler to measure and to draw lines exactly.	Lesson 9 of Term 1, 90 minutes. Objective: measure and draw lines in whole centimetres with a ruler. Time plan: opening 0 to 4, warm-up 4 to 10, sneak-peek 10 to 14, explore 14 to 28, mini-lesson 28 to 38, brain break 38 to 42, workbook 42 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Nobody is told which page they should be on. Tell the group the lesson is recorded (the tutor, the screen and any voice that is not muted) and that a child may keep the microphone and camera off.
2	4–10	**Warm-up: measure with clips** s-rem1	Lay clips along your pencil, end to end. / No gaps. No overlaps. / Count the clips. / WHAT IF?: / Your friend uses bigger clips. Will the number be the same?	Children lay paper clips (or any same-size things) along a pencil and count them. This is a spiral review of measuring in equal steps (holiday topic) and it lets a new child join: they see how it is done before they try. Show on camera with big clips and then small clips: the same pencil gives two different numbers, so we need one unit that everybody shares. Say centimetre only at the end of the part.
3	4–10	**Find 0 on your ruler** s-rem2	Put your finger on the 0 mark. Count along the marks. / [picture: number_line] / Say the numbers with me	Every child puts a finger on the 0 mark of their own ruler and counts along with you: 0 to 10, then on to 12. The strip on screen shows the numbers only, the real ruler is the one the child holds. Quick questions: the mark after 4 is 5, the mark before 10 is 9, 12 comes after 10. Rulers with a gap before the 0 are normal: we start at the 0 mark, not at the edge.
4	4–10	**One centimetre** s-rem3	The gap between two marks is 1 cm. / [picture: number_line] / Slide your finger from 0 to 5. How many gaps? How many cm?	A centimetre (cm) is the gap from one mark to the next. From 0 to 5 is five gaps, so 5 cm. The 0 is where we start: it is not a centimetre. Answers to the quick questions: 0 to 8 is 8 cm, 0 to 10 is 10 cm. If a child says 6 for 0 to 5, they counted the marks: ask them to count the jumps.
5	10–14	**Sneak-peek puzzles** s-sneak	Ideas we have not learnt yet. A guess is a great start. Nobody is marked. / PUZZLE 1: anyone can try: / Draw a dot. / Use your ruler. / Draw a line from the dot, exactly 3 cm long. / PUZZLE 2: think: / Ali puts one end of a pencil at the 1 mark of his ruler. / The other end is at the 9 mark. / Ali says it is 9 cm long. Is he right? Write yes or no. Then write why.	Puzzle 1 (a 3 cm line) is an access question: any child with a ruler can try it. Puzzle 2 is a thinking question on the lesson idea: where the pencil starts matters. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Everyone holds the paper to the camera once, at the same moment, after puzzle 2; there is no private chat and no board in front of the group. Answers: a line 3 cm long; no, 8 cm (the pencil starts at the 1 mark). Do not reveal Puzzle 2 until Talk time. Do not write scores or keep a list of who found it easy.
6	14–28	**Explore: guess, then measure** s-explore1	Find 3 things: a pencil, a spoon, a rubber or a toy. / Guess how many cm long each one is. / Put one end at the 0 mark. Read the mark at the other end. / Say: my ___ is about ___ cm. / WHAT IF?: / The end is between two marks. Say: about ___ cm.	Round 1, about 6 minutes. Objects first, little writing. Children guess, then measure three things from home. Ask for a thumbs up when each thing is measured; listen for the 0 mark in the answers. The what-if card is open to everyone: either mark next to the end is fine today, choosing the closer mark is taught as a Think harder idea in the mini-lesson. A child with a 15 cm ruler measures shorter things.

7	14–28	**Draw it!** s-explore2	Put a dot where the line starts. / Put the 0 mark on the dot. / Make a small mark at the end. / Join them with a straight line. / 3 cm 8 cm 12 cm	Round 2, about 5 minutes. Call each length and let children draw it. Children look at their own lines and show a thumbs up; nobody holds paper to the camera. Then ask about Ravi, who started at the edge of the ruler and not at the 0 mark: his line is a little longer than 8 cm. Answers: any line drawn with the 0 mark at the start and the end mark at 3, 8 and 12.
8	14–28	**What did we notice?** s-explore3	Where do we start measuring? / Why do we all use centimetres? / Why is a ruler better than clips?	Take one or two answers from volunteers. Key ideas: start at the 0 mark, not at the edge; read the mark at the other end; when the end is between two marks say about and name a mark next to it (choosing the closer mark comes in the mini-lesson as a Think harder idea); everybody uses the same unit so everybody gets the same answer (the clips gave different numbers). Repeat a child's answer in your own words so the recording picks it up.
9	28–38	**The parts of a ruler** s-m1	[picture: number_line] / REMEMBER: / 0 is where we start. / The gap from one mark to the next is 1 cm.	Hold your real ruler to the camera next to this slide. The numbers sit on the marks; the centimetre is the gap between two marks. Ask: how many centimetres from the 0 mark to the 6 mark? Answer: 6. Say the word mark every time, and say gap or jump for the centimetre.
10	28–38	**Measure: three steps** s-m2	[picture: ruler] / Put one end of the thing at the 0 mark. / Keep the ruler straight along the thing. / Read the mark at the other end. That is the length in cm. / Look from straight above	Point to the pencil on the ruler picture: one end exactly on the 0 mark, the tip on the 7 mark. Then say the three steps as you show them with a real pencil if you wish. Say each step as you do it. Read the mark at the tip aloud, with the unit: 14 cm, for example. Looking from straight above matters: from the side the marks seem to shift. Do not ask a child to demonstrate yet.
11	28–38	**Draw a line 7 cm long** s-m3	Dot where the line starts. / The 0 mark on the dot. / Small mark at the 7. / Join with a straight line. / Check: 0 at the start, 7 at the end. / [picture: ruler]	Draw the line on paper under your camera while you say the five steps. Hold the ruler still with one hand and draw along the edge with the other. At the end, check aloud: the 0 mark was on the dot and the 7 mark is at the other end. A line that is a little long is the most common slip: it usually means the line started at the edge of the ruler.
12	28–38	**Between two marks** s-m4	[picture: ruler] / The end is between the 8 mark and the 9 mark. It is closer to the 9 mark. / 8 9 / NEAREST CENTIMETRE: / Choose the closer mark. This one is about 9 cm.	THINK HARDER PREVIEW: Start here never needs this idea. Point to the picture: the tip falls between two marks. When the tip is clearly closer to one mark, choose that mark: that is the nearest centimetre. Ask the second example: between 12 and 13, closer to 12: answer 12 cm. Today every example is clearly closer to one mark; do not set one exactly halfway.
13	28–38	**A line that starts at 3** s-m5	[picture: number_line] / The line starts at 3 and ends at 8. Count the jumps. 5 jumps. The line is 5 cm long. / 8 take away 3 is 5	A line from the 3 mark to the 8 mark: the numbers show six marks, but only five jumps between them. Count the jumps aloud: 1, 2, 3, 4, 5. The length is the end mark take away the start mark: 8 take away 3 is 5. It is not 8, because the first 3 cm of the ruler are not part of the line. Ask: why is it 5 cm and not 8 cm? Link back to puzzle 2 only after Talk time.
14	28–38	**Your turn** s-m6	Write on your paper. About 2 minutes / DRAW: / Draw a line 5 cm long. / WORK OUT: / A line goes from the 2 mark to the 9 mark. / How long is it?	Each child draws and writes on their own paper. Wait until most children have stopped writing. Do not ask children to hold boards up in front of the group. If a child finishes early, they may check the line by measuring it again, starting at a different mark.
15	28–38	**Check: your turn** s-m6a	DRAW: / The line starts at the 0 mark and ends at the 5 mark. / WORK OUT: / 9 take away 2 is 7. / 7 cm	Read each answer aloud so a child watching the recording can mark their own work. The 5 cm line: 0 at the start, 5 at the end. The second line is 7 cm: count 7 jumps from 2 to 9, or take 2 from 9. If a child wrote 9, ask them to count the jumps from the 2.

16	38–42	**Jump the tens, hop the ones** s-break	Pretend a big jump is 10. A small hop is 1. / 14 23 30 8 / 14 is 1 big jump and 4 small hops. Sip some water.	Four minutes. Stand up on camera. Call each number and have children jump the tens and hop the ones; count the jumps and hops aloud. No equipment. Keep it gentle for small spaces. The big jump for ten links to bundles of ten from Lesson 1, but do not teach that link now.
17	42–70	**Workbook page 1: Start here** s-ws1	Questions 1 to 3. About 9 minutes. Use your ruler / questions S1, S2, S3	Read questions 1 to 3 aloud. Children work on their own; the tutor looks around the gallery and asks how do you know. Wait 10 seconds before helping. For question 3 a child may measure any short thing that fits on the ruler: the answer depends on the thing, so a grown-up checks it. After question 3 any child may choose to carry on or try Think harder.
18	42–70	**Check page 1** s-ws1ans	answers S1, S2, S3	Read each answer aloud. Children mark their own page in a different colour. Never ask a child to show a wrong answer to the group. Question 3 depends on the thing measured: a guess close to the measure is fine, and a whole number next to the tip is fine.
19	42–70	**Workbook page 2: Start here** s-ws2	Questions 4 to 6. About 8 minutes. Use your ruler / questions S4, S5, S6	Read question 4 aloud and show where the ruler picture is. Read question 6 aloud: it is a spot-the-mistake with a ruler picture. Children may check any answer by laying their own ruler next to the picture (it is printed at real size).
20	42–70	**Check page 2** s-ws2ans	answers S4, S5, S6	Read each answer aloud so a child watching the recording can mark their own work. Question 6: no, her line ends at the 7 mark so it is 7 cm; she should draw on to the 8 mark. Never ask a child to show a wrong answer to the group.
21	42–70	**Choose your next page** s-choice	Page | What it is / Think harder (page 3) | Our idea in a new setting / Go further (page 4) | Bigger puzzles / Investigation (page 5) | Does the start mark matter? / Check | Finish and check your own work / Everyone may choose. You can swap at any time.	Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 5 minutes.
22	70–76	**Talk and explain** s-talk	MEI AND BEN MEASURE THE SAME PENCIL: / Mei says 15 cm. / Ben says 16 cm. / Why might their answers be different? How can we check? / I think … because …	One minute of talk to the grown-up next to you, or to your screen. Then volunteers first; a pass is always fine; never call a child by name to answer. Model answer: the tip may be between the 15 and the 16 mark and the two children chose different marks, or one child did not start at the 0 mark. We can check by measuring again with the 0 at the start. Repeat each child's answer in your own words for the recording.
23	70–76	**Back to the sneak-peek** s-sneakans	[picture: number_line] / No! The pencil starts at the 1 mark, so it is 8 cm long: 9 take away 1.	Reveal puzzle 2 now. The pencil starts at the 1 mark and ends at the 9 mark: eight jumps, so 8 cm. Say: some of you may have found a good reason. Do not read out anyone's earlier answer.
24	76–82	**Exit ticket** s-exit	On your own. No looking back. About 3 minutes / questions E1, E2, E3	Children answer on page 5. Read the three questions aloud. Nobody holds the ticket up to the camera: children check their own ticket on the next slide and show fingers for how many they got right.
25	76–82	**Check the exit ticket** s-exitans	answers E1, E2, E3	Read each answer aloud. Children check their own ticket in a different colour. Question 1: the line starts at 0 and ends at 6. Question 2: 15 take away 6 is 9. Question 3: 10 take away 3 is 7.

26	82–85	**Look-back** s-look	[look-back box]	Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, the grown-up may photograph the page and send it to the business WhatsApp after class; open the next lesson with one catch-up question for the whole group. Never a direct message. The Next time I will try row is optional; at the start of the next lesson you may say to the whole group: today you could try drawing it, using objects or checking your answer.
27	85–90	**Well done, inquirers!** s-end	Show a grown-up one thing you found out today. / NEXT LESSON: / Story Steps and Mi's Challenge Quest. Have ready: about 20 counters, your ruler, a pencil and paper, the Quest Log in its envelope (keep it closed) and a quest partner (a grown-up, or a brother or sister).	Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. If you ran late, finish any open item (the look-back). Do not start anything new. If the whole class is ahead, show the optional menu slide and use the Spare-time menu from the tutor guide.
28	optional	**Go further: how many cuts?** s-opt	A strip of paper is 10 cm long. You may fold it at the 1 mark, the 2 mark and so on, but not at the ends. / How many places can you fold? What if the strip is 20 cm long?	Optional, only if you are on time. Answer: 9 places for the 10 cm strip (the marks 1 to 9), and 19 places for the 20 cm strip. We cannot fold at the 0 mark or the 10 mark, because that is the end of the strip. Take two answers and move on.
29	optional	**If we have time: pick one** s-menu	Activity | What we do / Is it always true? | A line ends at the 9 mark. Is it always 9 cm long? / Make a puzzle | Draw a path with 3 straight pieces for your grown-up to measure / Teach it back | Show how to draw a line 9 cm long, in steps / Optional. Everyone may join.	Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum.

8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

Chapter	Part	You say (recording voice)	A child watching later
0:00	Opening	*If you are watching the recording: hello! Get a pencil, paper, your workbook and your ruler. Pause or rewind any time.*	Gather your ruler, workbook, pencil and paper. Press play when ready.
4:00	Remember warm-up	*Recording: join in! Lay clips along a pencil, find 0 on your ruler, and count the marks out loud. Pause if you need longer.*	Play along. Do each step with your own ruler before the video does.
10:00	Sneak-peek puzzles	*Recording: pause now and try both puzzles on your own paper. Guessing is fine. Press play for the next part.*	Pause. Try both puzzles on your own paper. A guess is fine.
14:00	Explore	*Recording: pause now. Guess, then measure three things. Then draw lines of 3, 8 and 12 centimetres. Press play when you have tried.*	Pause. Guess and measure three things, then draw each line with your ruler. Write what you find.
28:00	Mini-lesson	*Recording: keep watching and copy each example with your own ruler. Rewind if you miss a step.*	Watch and copy the examples with your ruler. Rewind any step you missed.
38:00	Brain break	*Recording: stand up and join in! Big jump for ten centimetres, small hop for one.*	Stand up and jump the tens and hop the ones.

42:00	Workbook	*Recording: pause now and do workbook page 1 with your ruler. When you finish a page, press play to check it.*	Pause. Do page 1, then play to check. Then page 2. Then choose a page to try.
70:00	Talk time	*Recording: pause. Tell a grown-up your answer to the talk question, and say why.*	Pause. Explain your answer to someone at home, with a reason.
76:00	Exit ticket	*Recording: pause and do the exit ticket on your own, with your ruler. Then press play to check it.*	Pause. Do the exit ticket alone, then play to check it.
82:00	Look-back	*Recording: pause and fill in your look-back box. You can pass.*	Pause. Fill in the look-back box, or pass.
85:00	Flex time and goodbye	*Recording: well done for finishing! Show a grown-up one thing you found out.*	Show a grown-up your workbook and tell them one thing you found out.

9. Timing check (model, not a stopwatch)

Part	Allocated	Script	Slow class	Flag
opening	4:00	3:03	3:31	ok
remember	6:00	5:06	6:00	OVER (slow class)
sneak	4:00	3:26	4:00	ok
explore	14:00	11:42	13:53	ok
mini	10:00	8:24	9:53	ok
break	4:00	3:20	3:58	ok
work	28:00	26:39	31:44	OVER (slow class)
talk	6:00	5:27	6:28	OVER (slow class)
exit	6:00	5:08	6:01	OVER (slow class)
look	3:00	2:34	3:02	OVER (slow class)
flex	5:00	1:11	1:21	ok
Total	90:00	76:01	89:52	

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 6 questions; a middle child needs about 8.5 min for page 1 and 8.5 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.